

Alan Pipko

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EDUCATION

University of Guelph

Bachelor of Computer Science

Expected Graduation: Apr 2030

Relevant Courses: Programming, Data Structures, Linear Algebra, Discrete Mathematics, Web Design

WORK EXPERIENCE

Software Developer

VIBRAINT AI

Richmond Hill, ON

July 2026 – Present

- Developing real-time data visualization features in **JavaFX** for a production **Java** desktop application, modernizing legacy charting components
- Building live, continuously updating graph rendering for high-frequency data in a performance-sensitive UI
- Configured a reproducible **Java/JavaFX** build environment across multiple Windows machines, resolving JDK/SDK toolchain conflicts to unblock development

Machine Learning Intern

LineChecks - Desjardins Portfolio Company

Toronto, ON

Mar 2026 – Apr 2026

- Trained a YOLO model in Python to observe people at tennis/pickleball courts for live wait times
- Built and labeled an image dataset of court activity to tune the model for lighting and angles
- Developed a Python pipeline converting live occupancy counts into wait-time estimates

PROJECTS

Eura | *Python, FastAPI, React, TypeScript, PostgreSQL*

- Contributed to Eura, an AI-powered math tutoring platform with an interactive whiteboard
- Implemented **Google OAuth2** authentication using the Authorization Code flow with CSRF protection
- Built a **JWT session system** with short-lived access tokens and long-lived refresh tokens
- Added **email and password authentication** with bcrypt hashing alongside the OAuth flow
- Scoped per-user **localStorage** so each account maintains its own isolated canvas data

SnipSmart | *JavaScript, React, JSON*

- Utilized **Chrome Extension APIs** for on-screen capture and background processing
- Implemented **OCR-based text detection** to select text from captured screen snippets
- Added multilingual **text translation** support for extracted content
- Designed a lightweight **UI** to quickly copy and manage detected text with minimal user interaction

CERTIFICATIONS

IBM Cybersecurity Fundamentals | *Issued by IBM SkillsBuild*

May 2026

- Credential ID: 00b1ee3b-762e-4823-882f-5b4c71225cb8

TECHNICAL SKILLS

Languages: C, C#, Java, Python, SQL, JavaScript, HTML/CSS

Frameworks & Libraries: React, JavaFX, Next.js, YOLO, PostgreSQL, MonoGame, Tkinter, Pynput

Developer Tools: Git, Valgrind, Makefile, Linux Server, Render, Docker, Wireshark, Vite, Figma

LANGUAGES

English (Fluent) | French (Fluent) | Russian (Fluent)